

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon AWS PHL, PA -- station K12N, America/New_York
Building OSM 1334234458
Amazon AWS PHL
Facade gap not applicable -- one building, no second facade
Weather record 41,884 real hourly records from K12N
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-06-18 (chatter)
Plant limit 21.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 1 mode change. That is 7 more than the reactive incumbent, at 0 unsafe hours against its 1. 9 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 11 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 4 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
9 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	19.715	21.0	19.440	0.993
01	mechanical	yes	switch budget	20.275	21.0	20.560	0.889
02	mechanical	yes	switch budget	20.835	21.0	21.670	0.818
03	mechanical	yes	switch budget	20.000	21.0	20.000	0.739
04	mechanical	yes	switch budget	19.725	21.0	18.330	0.679
05	mechanical	yes	switch budget	19.725	21.0	17.220	0.626
06	mechanical	yes	switch budget	19.720	21.0	18.330	0.683
07	mechanical	yes	switch budget	20.275	21.0	18.330	1.455

08	mechanical	yes	switch budget	20.835	21.0	17.780	1.852
09	mechanical	no	dry-bulb	22.775	21.0	17.780	2.484
10	mechanical	no	dry-bulb	22.220	21.0	16.670	2.292
11	mechanical	no	dry-bulb	21.670	21.0	17.780	1.831
12	mechanical	no	dry-bulb	21.390	21.0	18.890	1.445
13	FREE	yes	-	19.170	21.0	17.220	1.001
14	FREE	yes	-	20.555	21.0	19.440	1.108
15	FREE	yes	-	19.445	21.0	17.220	0.970
16	FREE	yes	-	17.780	21.0	16.670	0.992
17	FREE	yes	-	18.055	21.0	16.110	1.046
18	FREE	yes	-	16.945	21.0	16.670	1.384
19	FREE	yes	-	16.395	21.0	16.670	1.525
20	FREE	yes	-	15.275	21.0	14.440	1.870
21	FREE	yes	-	15.280	21.0	13.890	1.570
22	FREE	yes	-	15.000	21.0	13.330	1.165
23	FREE	yes	-	13.610	21.0	12.780	0.862

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.715 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.275 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.835 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.000 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.725 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.725 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.720 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.275 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.835 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.775 C against a 21.0 C limit, so it fails by 1.775 C. It would take a limit 1.775 C higher, or a bound 1.775 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 21.0 C limit, so it fails by 1.220 C. It would take a limit 1.220 C higher, or a bound 1.220 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.670 C against a 21.0 C limit, so it fails by 0.670 C. It would take a limit 0.670 C higher, or a bound 0.670 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.390 C against a 21.0 C limit, so it fails by 0.390 C. It would take a limit 0.390 C higher, or a bound 0.390 C tighter, to change this hour.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.170 C, which is 1.830 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.001 C, and the margin is measured, not chosen: 1.001 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.555 C, which is 0.445 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.108 C, and the margin is measured, not chosen: 1.108 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.445 C, which is 1.555 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.970 C, and the margin is measured, not chosen: 0.970 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.780 C, which is 3.220 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.992 C, and the margin is measured, not chosen: 0.992 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.055 C, which is 2.945 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.045 C, and the margin is measured, not chosen: 1.045 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.945 C, which is 4.055 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.384 C, and the margin is measured, not chosen: 1.384 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.395 C, which is 4.605 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.525 C, and the margin is measured, not chosen: 1.525 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.275 C, which is 5.725 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.870 C, and the margin is measured, not chosen: 1.870 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.280 C, which is 5.720 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.570 C, and the margin is measured, not chosen: 1.570 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.000 C, which is 6.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.165 C, and the margin is measured, not chosen: 1.165 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.610 C, which is 7.390 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.862 C, and the margin is measured, not chosen: 0.862 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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